

A2 Economics

Accelerator!

National income and Investment



Zarak
Mishtaq

The accelerator: The theory focuses on induced investment and emphasises the volatility of investment. It states that investment depends on the rate of changes in income (and therefore consumer demand) and that a change in GDP will cause a greater proportionate change in investment. If a \$1 million increase in GDP causes induced investment to rise by \$3 million, the accelerator coefficient is said to be 3.

If GDP is rising, but at a constant rate, induced investment will not change. This is because firms can continue to buy the same number of machines each year to expand capacity. However, a change in the rate of growth of income can have a very significant influence on investment.

Table provides an example

year	consumer demand	machines at start of period	No. of machines required	Replacement investment	Induced Investment	Total Investment
1	800	8	8	1	0	1
2	1000	8	10	1	2	3
3	1600	10	16	1	6	7
4	1800	16	18	1	2	3
5	1800	18	18	1	0	1
6	1800	18	17	0	0	0

It is assumed that the firm starts the period with 8 machines, that one machine wears out each year and that each machine can produce 100 units of output per year.

The table shows that when demand for consumer goods rises by 25% (800 to 1000) in the second year, demand for capital goods rises by 200% (from 1 to 3). When the rate of growth in demand for consumer goods slows in year 4, demand for capital goods falls. By year 6, investment falls to zero with the worn-torn machine not being replaced, and so productive capacity is reduced.

However, an increase in demand for consumer goods does not always result in a greater percentage change in demand for capital goods. For instance, firms will not buy more capital goods if they have spare capacity or if they do not expect the rise in consumer demand to last. It may also not be possible for firms to buy many capital goods if the capital goods industries are working close to capacity. In addition with advances in technology, the capital-output ratio may change, with fewer machines being needed to produce given output.

Question

Q Increase in investment increases real national income or
Increase in income increases investment? 20

Ans:- Explain the concept of Investment

$\hat{I}$, autonomous, explain it

$\hat{I} \uparrow$, multiplier process, numerical e.g. $RNI \uparrow$
\$100, 5, $mpc = 0.8$, $RNI = \$500$

Define Accelerator

Use the table to explain the concept

conclusion:- Relationship between multiplier and accelerator

